to accelerate the workflow) using blockchain platforms. Typically, this can be designed with a permissioned, private blockchain platform to avoid the confusion between suppliers and customers when processing orders.

Sometimes, for rare and critical medical supplies, we need to design a hybrid or public blockchain platform so that new suppliers can be added quickly to it in order to process *ad hoc* supplies when there is a high demand or low availability of stocks with existing suppliers. For example, hand sanitisers and N95 masks had a huge demand during the initial three months of the pandemic, which required the inclusion of new suppliers for fulfilling this quickly.

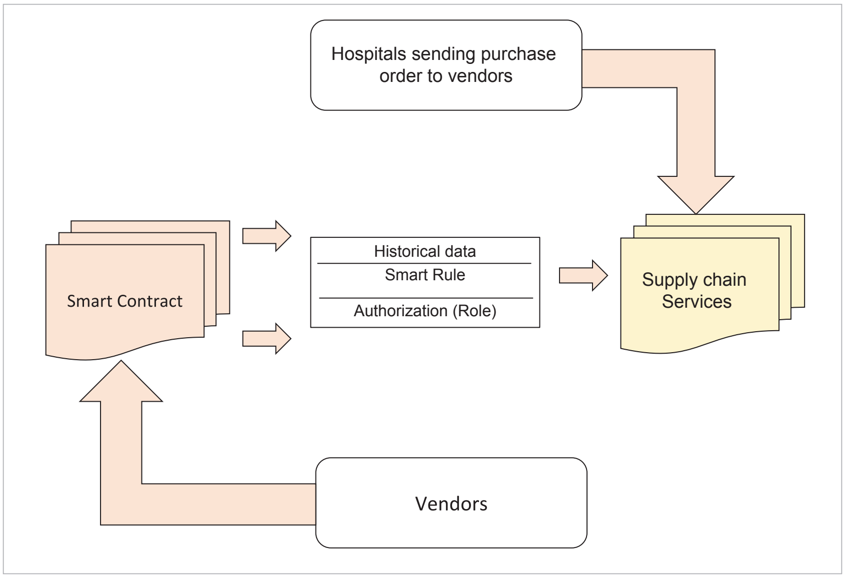

Figure 5: Blockchain based medical supply chain workflow

How challenges to blockchain healthcare solutions can be addressed

According to Gartner, since the blockchain platform offers features like decentralised ledger copies, encryption and certification authority for data access, distributed roles and governance, token based transaction access and immutability of records, there are three kinds of blockchain based solutions that can be developed for the healthcare and medical supply chain.

Blockchain inspired solution: In this, a semi-automated blockchain solution is created to address distributed data sources, data encryption during rest and in transit, and immutability of transaction records.

Blockchain-complete solution: This is a blockchain based solution that addresses token-based transaction handling for higher security, decentralised ledger copies across the network, a distributed data source, data encryption during rest and in transition, and immutability of transaction records.

Enhanced blockchain solution: This can be an extension of the blockchain-complete solution by providing advanced technology support like AI/ML based analytics services for smart contract handling, IoT for data management, and self-sovereign identity (SSI) for improved security features.

The impact and future of blockchain solutions in the healthcare industry

With the power of cloud based blockchain platforms like AWS blockchain, Azure blockchain services and BlockCloud platforms, issues like counterfeit drugs can be handled. When new drugs (medicines, vaccines, medical accessories) are designed, the supplier needs to add the product portfolio as a block to the blockchain network in order to make it available in the market and add it to the supply chain. Future use cases of the blockchain include handling relief operations during man-made or natural disasters, and remote consultations with a team of doctors who practice in different regions of the world.

Cloud based, agile, scalable and secure platforms that integrate quickly with multi-technology solutions are some of the features that will make blockchain solutions popular in the healthcare industry in the near future. END

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